

2. (a) Explain what is meant by 'centripetal acceleration'. [2]

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(b) The Earth has a radius of approximately 6370 km and rotates about its axis once each day. A student stands at the equator.

(i) Show that the angular velocity of the student is approximately $7 \times 10^{-5} \text{ rad s}^{-1}$. [2]

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(ii) Determine the centripetal acceleration of the student. [2]

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(iii) I. Determine the speed of the student around the circular path. [2]

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II. The student says, "this is a very high speed so my answer must be incorrect as I would feel the effects of this speed." Discuss whether or not the student is correct. [2]

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